Statistical Tables Cheat Sheet

Quick Reference for Critical Values

Z-Table (Standard Normal)

Common Z-Values

Confidence Level	α	α/2	z*
90%	0.10	0.05	1.645
95%	0.05	0.025	1.96
98%	0.02	0.01	2.33
99%	0.01	0.005	2.576

Percentile Z-Scores

Percentile	Z
90th	1.28
95th	1.645
97.5th	1.96
99th	2.33

Area Under Curve

Z	Area Left	Area Right
1.0	0.8413	0.1587
1.5	0.9332	0.0668
2.0	0.9772	0.0228
2.5	0.9938	0.0062
3.0	0.9987	0.0013

T-Table (Student's t)

Critical t-Values (Two-Tailed)

df	90%	95%	98%	99%
5	2.015	2.571	3.365	4.032
10	1.812	2.228	2.764	3.169
15	1.753	2.131	2.602	2.947
20	1.725	2.086	2.528	2.845
25	1.708	2.060	2.485	2.787
30	1.697	2.042	2.457	2.750
40	1.684	2.021	2.423	2.704
60	1.671	2.000	2.390	2.660
100	1.660	1.984	2.364	2.626
∞	1.645	1.960	2.326	2.576

One-Tailed t-Values

df	α=0.05	α=0.025	α=0.01
10	1.812	2.228	2.764
20	1.725	2.086	2.528
30	1.697	2.042	2.457

Chi-Square Table

Critical χ^2 Values ($\alpha = 0.05$)

df	χ^2
1	3.841
2	5.991
3	7.815
4	9.488
5	11.071
10	18.307
15	24.996
20	31.410

Quick Formulas

Sample Size Calculations

```
Mean: n = (z*σ/ME)²
Proportion: n = p̂(1-p̂)(z*/ME)²
```

Standard Error

```
Mean: SE = σ/√n or s/√n
Proportion: SE = √[p̂(1-p̂)/n]
```

Confidence Intervals

```
Mean (σ known): x̄ ± z*(σ/√n)
Mean (σ unknown): x̄ ± t*(s/√n)
Proportion: p̂ ± z*√[p̂(1-p̂)/n]
```

Test Statistics

```
Z-test: z = (x̄ - μ₀)/(σ/√n)
T-test: t = (x̄ - μ₀)/(s/√n)
```

Interpretation Guide

P-Value Interpretation

P-value	Strength of Evidence
< 0.01	Very strong
0.01-0.05	Strong
0.05-0.10	Moderate
> 0.10	Weak/None

Effect Size (Cohen's d)

$\|d\|$	Interpretation
0.2	Small
0.5	Medium
0.8	Large

Keep this reference handy for homework and exams!

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